Extracellular water characterizes sex-specific differences in Autism Spectrum Disorder

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Introduction

Autism spectrum disorder (ASD) is a complex, multifaceted disorder with a significant sex bias in both diagnosis rate and behavioral symptom profile[1]. Previous studies have found structural differences in neuronal white matter (WM) with ASD being associated with lower FA and higher diffusivity[2]. Studies on sex differences have suggested that ASD males and females may lack sex-specific differences seen in typically developing individuals[3]. This study aims to characterize sex-specific differences in ASD using advanced models of brain microstructure including a metric of extracellular water.

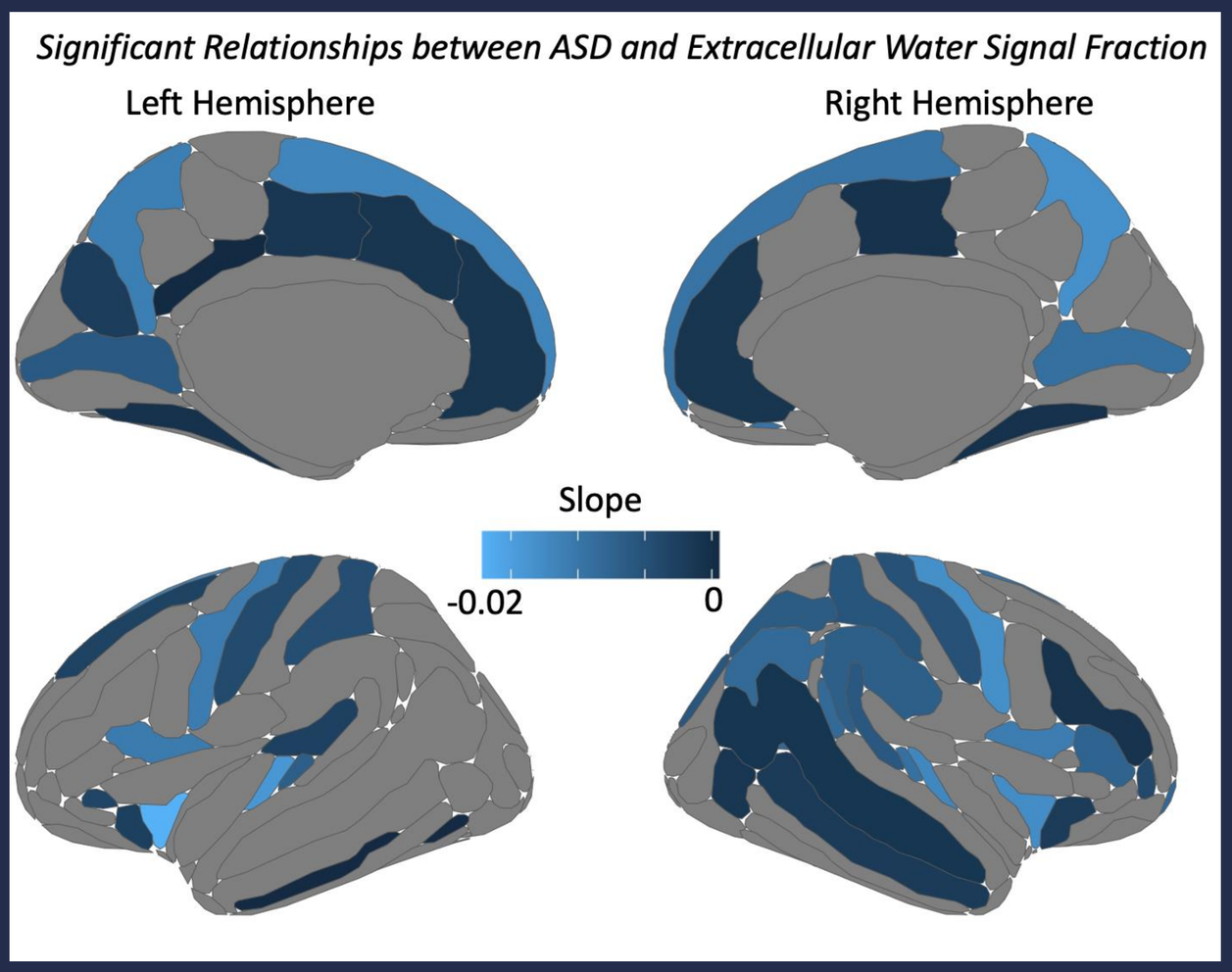

Figure 1: Map of ROIs that were significant at the p<0.05 level after multiple comparison correction. Colored by slope of relationship, with ASD individuals having greater extracellular water signal fraction in each ROI.

Methods

Participants - 290 (144 female, mean age = 154.4 months ±36.15 S.D.) subjects from the NIH-sponsored Autism Centers for Excellence Wave 1 were included in this study. Study cohort included 151 individuals diagnosed with ASD (75 female, mean age = 151.4 months ±35.22 S.D.) and 139 neurotypical controls (69 female, mean age = 158.0 months ±36.99 S.D.).

Image Acquisition and Analysis - Diffusion, T1-weighted, and T2-weighted images were acquired from each subject. Diffusion images were acquired with an isotropic voxel size of 2x2x2mm, 64 non-colinear gradient directions at b=1000 s/mm^2 and 1 b=0. T1-weighted MPRAGE images with an FOV of 176x256x256mm and an isotropic voxel size of 1x1x1mm, TE=3.3; T2-weighted images were acquired with an FOV of 128x128x34 with a voxel size of 1.5x1.5x4mm, TE=35. As described in Newman et al., (2020), diffusion images were preprocessed and 3-Tissue Constrained Spherical Deconvolution used to calculate the fraction of signal arising from each of 3 compartments: an intracellular anisotropic, intracellular isotropic, and extracellular isotropic freely diffusing water compartment. T1w and T2w images were processed as described in the MICA-MNI pipeline and the T1/T2 ratio calculated on a voxelwise basis. All imaging metrics were averaged across each of 214 ROIs taken from the JHU-ICBM WM atlas (48 ROIs) and the Destreaux Cortical Atlas (164 ROIs). Additionally 2 composite ROIs were included, one of all 48 JHU ROIs and one of 150 neocortical regions from the Destreaux Atlas.

Statistical Approach - General linear models were used to test the relationship between each microstructural metric (intracellular anisotropic, intracellular isotropic, extracellular free water, and T1/T2 ratio) on diagnosis and sex differences. All models featured subject age, IQ, scanner/site, and intracranial volume as control terms. All sets of microstructural results were correcting for multiple comparisons using the Benjamini & Hochberg (1995) method across all 214 ROIs. Exploratory graph analysis was applied to significant ROIs.

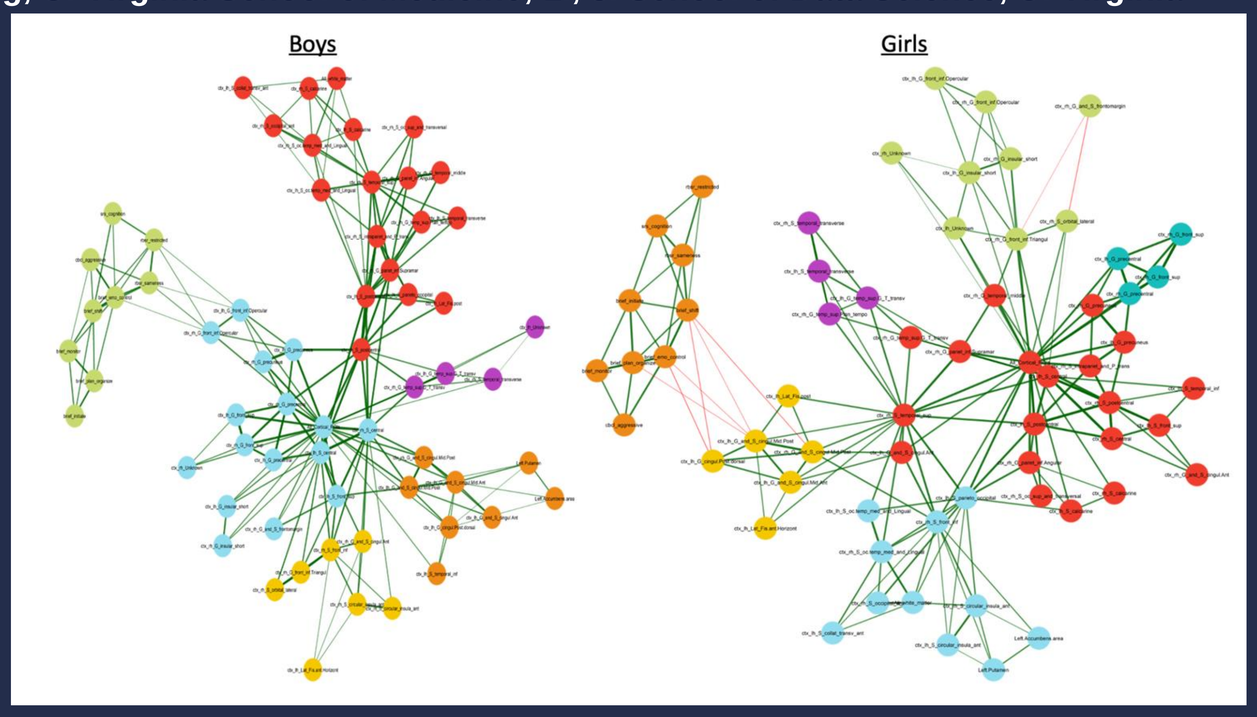

Figure 2: Exploratory Graph Analysis Network of ROIs with significant differences in extracellular water signal fraction examined alongside behavioral metrics in ASD boys and ASD girls. Cognitive behavioral deficits most closely vary with opercular cortex extracellular water in boys but in girls have a negative association with the cingulum. Network colors are arbitrarily assigned.

Results

54 ROIs were found to have significantly greater extracellular water signal fraction in individuals with ASD compared to controls (Fig. 1), including in both composite ROIs covering the WM and neocortex. No other microstructure metric survived multiple comparison correction in any ROI. However, no relationship was found between the microstructural metrics and the behavioral metrics; the closest relationship was between intracellular axonal and T1/T2 ratio measured in the bilateral transverse frontopolar gyri and sulci and the CBCL Aggression subscale, which had a p<0.01 uncorrected but was p<0.0535 after accounting for multiple comparisons.

To test the relationship between behavioral metrics in each sex, we performed exploratory graph analysis on the extracellular water results from ROIs that were significantly related to ASD diagnosis (Fig. 2). This showed that behavioral metrics most closely vary with opercular cortex extracellular water in boys but in girls have a negative association with the cingulum. From this we hypothesized that ASD girls may show deficits in the WM areas while ASD boys may show deficits in neocortex compared to controls. Separating the models by atlas, sex, and diagnosis resulted in no significant differences between ASD boys and ASD girls; however, ASD girls had significantly reduced extracellular water in 5 JHU ROIs compared to controls (including the cingulum), while ASD boys had significantly greater extracellular water in only the composite JHU ROI compared to controls. Conversely, ASD girls had no significant differences from controls in any microstructural metric in any Destreaux Cortical atlas ROI, while ASD boys had significantly greater extracellular water in the anterior occipital lobe, an ROI previously implicated in ASD.

Conclusion

This study shows widespread significant differences between ASD and neurotypical controls, while also highlighting significant sex-dependent differences in behavioral deficits and brain structure compared to same-sex controls.

References

1. Werling, D. M., & Geschwind, D. H. (2013). Sex differences in autism spectrum disorders. Current opinion in neurology, 26(2), 146.
2. Dimond, D., Schuetze, M., Smith, R.E., Dhollander, T., Cho, I., Vinette, S., … & Bray, S. (2019). Reduced white matter fiber density in autism spectrum disorder. Cerebral Cortex, 29, 1778–1788.
3. Peterson, B. S., Liu, J., Dantec, L., Newman, C., Sawardekar, S., Goh, S., & Bansal, R. (2022). Using tissue microstructure and multimodal MRI to parse the phenotypic heterogeneity and cellular basis of autism spectrum disorder. Journal of Child Psychology and Psychiatry, 63(8), 855-870.
4. Newman, B. T., Dhollander, T., Reynier, K. A., Panzer, M. B., & Druzgal, T. J. (2020). Test–retest reliability and long-term stability of three-tissue constrained spherical deconvolution methods for analyzing diffusion MRI data. Magnetic resonance in medicine, 84(4), 2161-2173.

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